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# MXene–PVA composite for arsenic removal from industrial wastewater: a combined DFT and experimental study

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Arsenic pollution in industrial effluent presents significant environmental and health hazards worldwide. Although  $\text{Ti}_3\text{C}_2\text{T}_x$  MXene demonstrates potential for arsenic adsorption, its actual use is constrained by structural instability and aggregation in aqueous conditions. This research combines density functional theory (DFT) computations with experimental verification to create MXene–polyvinyl alcohol (PVA) composites for efficient arsenic elimination. DFT simulations indicated an increased As(v) adsorption energy of  $-2.58$  eV due to synergistic hydrogen bonding between the hydroxyl groups of PVA and the surface terminations of MXene. The synthesized MXene–PVA composite (20 wt% MXene) attained a maximum adsorption capacity of  $135.2 \text{ mg g}^{-1}$  for As(v), adhering to pseudo-second-order kinetics and the Langmuir isotherm model. The composite exhibited 85% selectivity in the presence of competing ions ( $\text{PO}_4^{3-}$ ,  $\text{SO}_4^{2-}$ ) and preserved 85% capacity following eight regeneration cycles. Actual wastewater treatment from electroplating facilities lowered arsenic levels from  $8.2 \text{ mg L}^{-1}$  to  $0.008 \text{ mg L}^{-1}$ , thereby complying with WHO requirements. XPS study verified that bidentate As–O–Ti complexation is the predominant process, corroborating DFT predictions. This study illustrates a feasible approach for industrial arsenic cleanup using an integrated computational–experimental design.

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## 1. Introduction

Arsenic contamination impacts more than 200 million individuals worldwide, with industrial processes like electroplating and mining releasing effluents with  $5\text{--}15 \text{ mg L}^{-1}$  of arsenic, significantly beyond the WHO's recommendations of  $0.01 \text{ mg L}^{-1}$ .<sup>1,2</sup> Traditional technologies such as coagulation–flocculation and membrane filtration are hindered by elevated costs, secondary contamination, and inadequate selectivity.<sup>3</sup> Adsorption technologies have benefits in terms of simplicity and cost-efficiency; nevertheless, conventional adsorbents such as activated carbon ( $18 \text{ mg g}^{-1}$ ) and zeolites ( $28 \text{ mg g}^{-1}$ ) exhibit restricted capacity.<sup>4,5</sup> Since their discovery in 2011, MXenes, which are two-dimensional transition metal carbides represented by the formula  $\text{M}_{(n+1)}\text{X}_n\text{T}_x$ , have emerged as promising adsorbents.<sup>6,7</sup>  $\text{Ti}_3\text{C}_2\text{T}_x$  has exceptional conductivity ( $6500 \text{ S cm}^{-1}$ ), extensive surface area ( $>400 \text{ m}^2 \text{ g}^{-1}$ ), and several functional groups ( $-\text{OH}$ ,  $-\text{O}$ ,  $-\text{F}$ ), offering multiple binding sites.<sup>8,9</sup> Prior theoretical analyses indicated that  $\text{Ti}_3\text{C}_2(\text{OH})_2$  attains an As(v) binding energy of  $-2.43$  eV with a charge transfer of  $0.34 \text{ e}^-$ .<sup>10</sup> Pristine MXene encounters significant challenges: (i) oxidation and degradation in aqueous environments, (ii) layer restacking that diminishes accessible surface area, and (iii) complex separation from treated water.<sup>11,12</sup>

Polyvinyl alcohol (PVA) has superior film-forming capabilities and mechanical strength due to hydrogen bonding networks, which provide structural support to inhibit MXene aggregation while adding hydroxyl groups for improved adsorption.<sup>13,14</sup> Molecular-level interactions at MXene–PVA interfaces and their impact on arsenic adsorption, despite their great potential, are inadequately comprehended. The existing literature is deficient in: (1) thorough DFT simulations forecasting MXene–polymer–arsenic interactions in aquatic settings, (2) methodical assessment of performance amid competing ions, and (3) verification utilizing actual industrial wastewater with intricate matrices.

This study overcomes these deficiencies using an integrated computational–experimental methodology. The objectives are: (1) utilizing DFT and *ab initio* molecular dynamics (AIMD) to clarify the adsorption mechanisms of As(v) on MXene–PVA interfaces, (2) synthesizing and characterizing optimized MXene–PVA composites, (3) assessing adsorption performance, including pH effects, kinetics, isotherms, and selectivity, (4) validating efficacy with genuine electroplating wastewater, and (5) developing a mechanistic understanding through XPS and spectroscopic analysis aligned with computational predictions.<sup>15–18</sup>

## 2. Materials and methods

### 2.1. Computational methodology

Density Functional Theory (DFT) computations were conducted utilizing the Vienna *Ab initio* Simulation Package (VASP 5.4.4)

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with the Perdew–Burke–Ernzerhof (PBE) functional and the projector augmented wave (PAW) method. The plane-wave kinetic energy cutoff was set at 500 eV, confirmed by convergence tests indicating energy fluctuations of less than 1 meV per atom. Brillouin zone sampling employed a  $3 \times 3 \times 1$  Monkhorst–Pack  $k$ -point mesh for optimization and a  $5 \times 5 \times 1$  mesh for electrical properties.<sup>19</sup> The MXene supercell comprised a  $4 \times 4 \times 1$   $\text{Ti}_3\text{C}_2(\text{OH})_2$  monolayer (48 Ti, 32 C, 32 OH groups, approximately  $15 \times 15$  Å dimensions) with a vacuum spacing of 20 Å. Hydroxyl termination was chosen because of its improved arsenic adsorption relative to –F and –O terminations.<sup>10</sup> PVA was represented as  $-(\text{CH}_2-\text{CHOH})_{10}-$  oligomer situated 2.5–3.5 Å above the MXene surface. Arsenic species were represented as  $\text{H}_2\text{AsO}_4^-$  for As(v), indicating the predominant form at pH 6–8.<sup>20</sup> Various initial layouts were evaluated with arsenic at atop, bridge, and hollow sites. Structural optimizations achieved convergence with forces below  $0.01 \text{ eV } \text{\AA}^{-1}$  and energy below  $10^{-5} \text{ eV}$  *via* the conjugate gradient technique. Dipole adjustments were implemented in the  $z$ -direction.<sup>21</sup> The adsorption energy ( $E_{\text{ads}}$ ) was computed as follows:

$$E_{\text{ads}} = E[\text{MXene-PVA-As}] - E[\text{MXene-PVA}] - E[\text{As(species)}] \quad (1)$$

Solvation effects were incorporated with VASPsol implicit solvation ( $\epsilon = 78.4$ ).<sup>22</sup> Bader charge analysis assessed charge transfer, while density of states (DOS) evaluated alterations in electronic structure following adsorption.<sup>23,24</sup>

AIMD simulations utilized the canonical (NVT) ensemble with a Nosé–Hoover thermostat at 298 K, with a time step of 1 fs and a duration of 20 ps. A reduced cutoff of 400 eV and  $\Gamma$ -point sampling were employed in accordance with established protocols for large-scale dynamics.<sup>25</sup> Explicit water, comprising 80  $\text{H}_2\text{O}$  molecules in a  $\sim 10$  Å layer with a density of  $1.0 \text{ g cm}^{-3}$ , enveloped the compound. Structural stability was evaluated using root-mean-square deviation (RMSD), radial distribution functions (RDFs), and hydrogen bond analysis ( $\text{O}\cdots\text{H} < 2.5$  Å,  $\text{O-H}\cdots\text{O} > 140^\circ$ ).<sup>26</sup>

## 2.2. Materials

$\text{Ti}_3\text{AlC}_2$  MAX phase (98%,  $<40$  µm, Carbon-Ukraine Ltd), LiF ( $\geq 99\%$ ), HCl (37%), PVA (Mw 89 000–98,000, 99% hydrolyzed),  $\text{Na}_2\text{HAsO}_4 \cdot 7\text{H}_2\text{O}$  ( $\geq 98\%$ ), and  $\text{NaAsO}_2$  ( $\geq 90\%$ ) were procured from Sigma-Aldrich and Merck. Deionized water (18.2 MΩ cm, Milli-Q) was utilized consistently.

## 2.3. MXene synthesis

$\text{Ti}_3\text{C}_2\text{T}_x$  was synthesized using the minimally intensive layer delamination (MILD) method.<sup>27</sup> LiF (1.0 g) was solubilized in 20 mL of 9 M HCl, after which  $\text{Ti}_3\text{AlC}_2$  (1.0 g) was included and agitated at 35 °C for 24 hours. The slurry was subjected to centrifugation with deionized water (3500 rpm, 5 minutes cycles) until a pH of approximately 6 was achieved. Sonication (200 W, 40 kHz, 1 h) in an argon environment facilitated the delamination of the layers. Following centrifugation (3500 rpm,

30 minutes), the dark-green supernatant containing MXene nanosheets was obtained and preserved at 4 °C under argon atmosphere.

## 2.4. Synthesis of MXene–PVA composite

PVA (2.0 g) was solubilized in deionized water (20 mL) at 90 °C for 2 hours, resulting in a 10 wt% solution. Upon cooling to 60 °C, a 10 mL dispersion of MXene ( $5 \text{ mg mL}^{-1}$ ) was incrementally introduced while stirring for 30 minutes. The mixture was poured into Petri dishes and dried at 50 °C for 24 hours to create films, or extruded into liquid nitrogen and lyophilized for 48 hours to produce beads. MXene loading was around 20 weight percent based on solid composition. Mechanical stability was attained *via* physical crosslinking during three freeze–thaw cycles ( $-20$  °C/25 °C, 12 hours each).<sup>28</sup>

## 2.5. Characterization

X-ray diffraction (XRD, Bruker D8 Advance, Cu  $K\alpha$ ,  $\lambda = 1.5406$  Å) was conducted over a range of  $5$ – $80^\circ$  ( $2\theta$ ) with a step size of  $0.02^\circ$ . FTIR (Thermo Nicolet iS50, ATR mode) was conducted over the range of  $4000$ – $400 \text{ cm}^{-1}$ , utilizing 64 scans at a resolution of  $4 \text{ cm}^{-1}$ . Structural characteristics were studied by Raman spectroscopy (Horiba LabRAM HR, 532 nm, 5 mW). SEM-EDS (JEOL JSM-7800F with Oxford X-MaxN 150) analyzed morphology and elemental distribution subsequent to Au/Pd coating (5 nm). X-ray photoelectron spectroscopy (XPS) was conducted using a Thermo ESCALAB 250Xi with Al  $K\alpha$  radiation at 1486.6 eV, analyzing chemical states with a 20 eV pass energy, calibrated to C 1s at 284.8 eV.  $\text{N}_2$  adsorption–desorption (Micromeritics ASAP 2020, 77 K) assessed the BET surface area and BJH pore distribution following degassing at 120 °C for 12 hours. TGA (TA Q500) employed a nitrogen environment ( $60 \text{ mL min}^{-1}$ ) with a heating rate of  $10^\circ\text{C min}^{-1}$  from 30 to 800 °C.

## 2.6. Batch adsorption experiments

Stock solutions of As(v) and As(III) were formulated from  $\text{Na}_2\text{HAsO}_4 \cdot 7\text{H}_2\text{O}$  and  $\text{NaAsO}_2$ . The pH was modified utilizing 0.1 M HCl/NaOH. All experiments were performed in duplicate. Impact of pH: Composite (20 mg) was introduced to an arsenic solution ( $40 \text{ mL}$ ,  $20 \text{ mg L}^{-1}$ ) at pH 3–11, agitated (150 rpm, 25 °C) for 24 hours. Following centrifugation (4000 rpm, 10 minutes) and filtering ( $0.45$  µm), arsenic was assessed by ICP-OES (PerkinElmer Optima 8000).

Kinetics: Composite (50 mg) in arsenic solution (100 mL,  $50 \text{ mg L}^{-1}$ , pH 7.0) was sampled at intervals (5–1440 min). Adsorption capacity  $q_t$  ( $\text{mg g}^{-1}$ ) was:

$$q_t = (C_0 - C_t)V/m \quad (2)$$

where  $q_t$  is adsorption capacity at time  $t$  ( $\text{mg g}^{-1}$ ),  $C_0$  is initial concentration ( $\text{mg L}^{-1}$ ),  $C_t$  is concentration at time  $t$  ( $\text{mg L}^{-1}$ ),  $V$  is solution volume (L), and  $m$  is adsorbent mass (g).

Data were fitted to pseudo-first-order:<sup>29</sup>

$$\ln(q_e - q_t) = \ln(q_e) - k_1 t \quad (3)$$



where  $q_e$  is equilibrium adsorption capacity ( $\text{mg g}^{-1}$ ),  $k_1$  is pseudo-first-order rate constant ( $\text{min}^{-1}$ ), and  $t$  is time (min).

Pseudo-second-order kinetic model:<sup>29</sup>

$$t/q_t = 1/(k_2 q_e^2) + t/q_e \quad (4)$$

where  $k_2$  is pseudo-second-order rate constant ( $\text{g mg}^{-1} \text{min}^{-1}$ ).

Isotherms: initial As(v) concentration varied ( $10\text{--}200 \text{ mg L}^{-1}$ , pH 7.0,  $25^\circ\text{C}$ ,  $0.5 \text{ g L}^{-1}$  dosage, 24 h). Data were analyzed using Langmuir and the Freundlich models:<sup>30</sup>

Langmuir:

$$q_e = (q_{\text{max}} K_L C_e)/(1 + K_L C_e) \quad (5)$$

where  $q_{\text{max}}$  is maximum adsorption capacity ( $\text{mg g}^{-1}$ ),  $K_L$  is Langmuir constant ( $\text{L mg}^{-1}$ ), and  $C_e$  is equilibrium concentration ( $\text{mg L}^{-1}$ ).

Freundlich:

$$q_e = K_f C_e^{(1/n)} \quad (6)$$

where  $K_f$  is Freundlich constant ( $(\text{mg g}^{-1}) (\text{L mg}^{-1})^{(1/n)}$ ) and  $n$  is Freundlich exponent (dimensionless).

Selectivity: mixed solutions contained As(v) ( $10 \text{ mg L}^{-1}$ ) and competing anions ( $\text{PO}_4^{3-}$ :  $10\text{--}100 \text{ mg L}^{-1}$ ;  $\text{SO}_4^{2-}$ ,  $\text{HCO}_3^-$ :  $100 \text{ mg L}^{-1}$ ) at pH 7.0. Distribution coefficient ( $K_d$ ,  $\text{mL g}^{-1}$ ) and selectivity coefficient ( $\alpha$ ) were:

$$K_d = [(C_0 - C_e)/C_e] \times (V/m) \quad (7)$$

where  $K_d$  is distribution coefficient ( $\text{mL g}^{-1}$ ).

$$\alpha = K_d(\text{As})/K_d(\text{interfering ion}) \quad (8)$$

where  $\alpha$  is the selectivity coefficient (dimensionless).

Regeneration: As-loaded composites were treated with  $0.1 \text{ M NaOH}$  ( $50 \text{ mL}$ ,  $150 \text{ rpm}$ ,  $4 \text{ h}$ ,  $25^\circ\text{C}$ ), washed to neutral pH, dried at  $60^\circ\text{C}$ , and reused for 8 cycles.

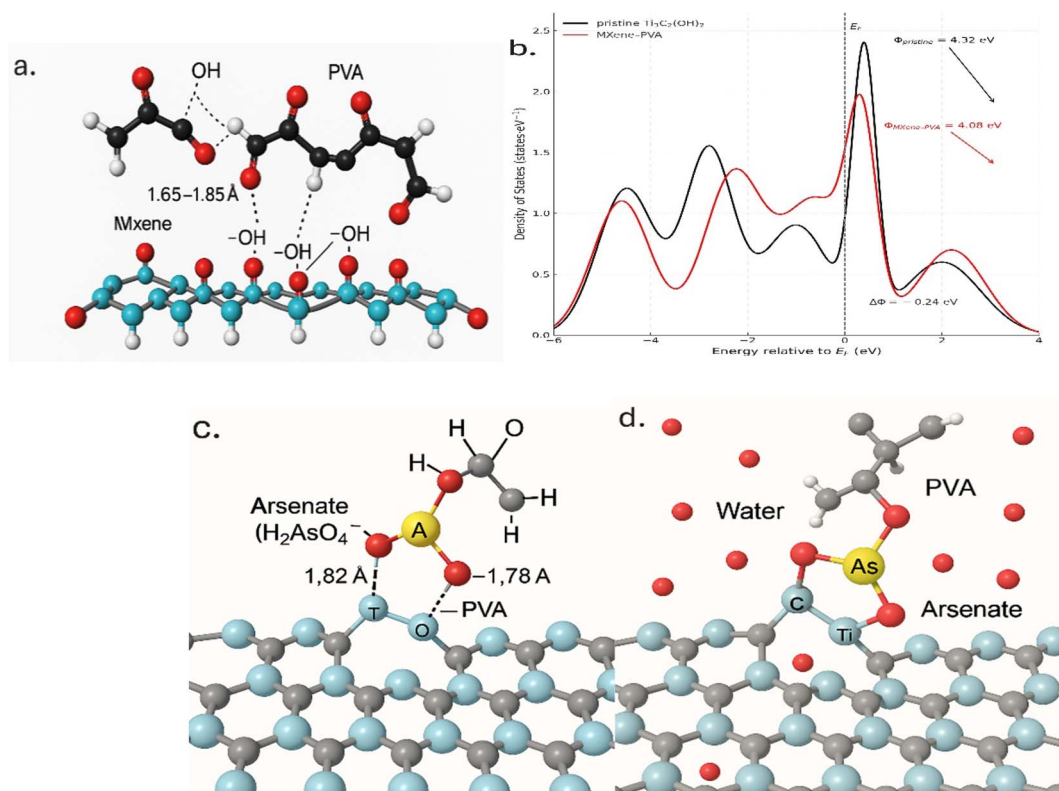
Low-concentration kinetics: to confirm applicability at wastewater-relevant concentrations, kinetic tests were conducted at a starting As(v) concentration of  $10 \text{ mg L}^{-1}$  (pH 7.0,  $25^\circ\text{C}$ ,  $0.5 \text{ g L}^{-1}$  dose). We used ICP-MS (with a detection limit of  $0.1 \mu\text{g L}^{-1}$ ) to check that the measurements were accurate at low concentrations.

Extended low-concentration isotherms: more isotherm data points were gathered at 1, 2, and  $5 \text{ mg L}^{-1}$  to fully cover the range of wastewater concentrations ( $5\text{--}15 \text{ mg L}^{-1}$  is normal for electroplating effluents).

Concentrations for investigations on selectivity and regeneration: As(v) =  $10 \text{ mg L}^{-1}$  was employed in selectivity tests, and the amounts of the competing ions were changed as shown in Table 9. Regeneration cycles used As(v) =  $50 \text{ mg L}^{-1}$  to get enough loading for valid capacity measurements following desorption.

## 2.7. Real wastewater treatment

Wastewater from an electroplating facility (Binh Duong, Vietnam) was collected, acidified to pH 2 with  $\text{HNO}_3$ , and stored at



**Fig. 1** Computational insights into MXene-PVA composite and arsenic adsorption. (a) Optimized structure of MXene-PVA interface. (b) Density of states (DOS) for pristine MXene and MXene-PVA. (c) As(v) adsorption configuration on MXene-PVA. (d) AIMD simulation results at 298 K.



4 °C. Samples were characterized for total arsenic, speciation (As(III)/As(V)), pH, conductivity, TDS, and major ions using IC (Dionex ICS-5000) and ICP-OES. Treatment used optimized conditions (pH 7.0, 1.5 g L<sup>-1</sup> dosage, 16 h, 25 °C). Treated water was analyzed by HPLC-ICP-MS (Agilent 1260-7900) for arsenic speciation.<sup>31</sup> TOC (Shimadzu TOC-L) assessed polymer leaching.

## 3. Results and discussion

### 3.1. Computational predictions

**3.1.1. MXene-PVA interface formation.** DFT calculations revealed extensive hydrogen bonding between PVA hydroxyl groups and MXene surface -OH terminations (Fig. 1a). O(PVA)⋯H-O(MXene) distances ranged 1.65–1.85 Å, characteristic of moderate-strength hydrogen bonds.<sup>32</sup> The binding energy of the PVA chain to MXene was -1.28 eV per repeating unit, indicating thermodynamically favorable adsorption. Bader analysis showed 0.08 e<sup>-</sup> transfer from PVA to MXene, suggesting partial ionic character complementing hydrogen bonding.

DOS analysis (Fig. 1b) revealed that PVA binding altered MXene electronic structure. Ti 3d states near the Fermi level

experienced redistribution, with work function decreasing from 4.32 eV (pristine MXene) to 4.08 eV (MXene-PVA), indicating enhanced electron-donating character.<sup>33</sup>

**3.1.2. Arsenic adsorption on MXene-PVA.** For As(V) adsorption (Fig. 1c), arsenate (H<sub>2</sub>AsO<sub>4</sub><sup>-</sup>) formed bidentate complexes with two Ti atoms (As-O-Ti bond lengths: 1.82, 1.85 Å). Simultaneously, hydrogen bonds formed between arsenate oxygen and PVA hydroxyl groups (O(As)⋯H-O(PVA) = 1.78 Å), creating a dual-binding mode. Adsorption energy was -2.58 eV with implicit solvation, representing 6% enhancement *versus* pristine MXene-OH (-2.43 eV).<sup>10</sup> Bader analysis revealed 0.38 e<sup>-</sup> transfer from As(V) to MXene-PVA composite (0.34 e<sup>-</sup> on MXene Ti/O atoms, 0.04 e<sup>-</sup> on PVA oxygen).

As(III) showed weaker binding (-1.85 eV) due to the uncharged nature of H<sub>3</sub>AsO<sub>3</sub> at neutral pH, primarily interacting through hydrogen bonding. Comparison across binding sites revealed hollow sites were 0.3–0.5 eV less favorable than atop or bridge sites, explaining incomplete surface Ti utilization.

**3.1.3. AIMD simulations.** AIMD simulations using explicit water molecules (298 K, 20 ps) exhibited the dynamic stability of the MXene-PVA-As(V) complex (Fig. 1d). The RMSD of arsenic locations stabilized at around 0.35 Å after 5 ps, with no

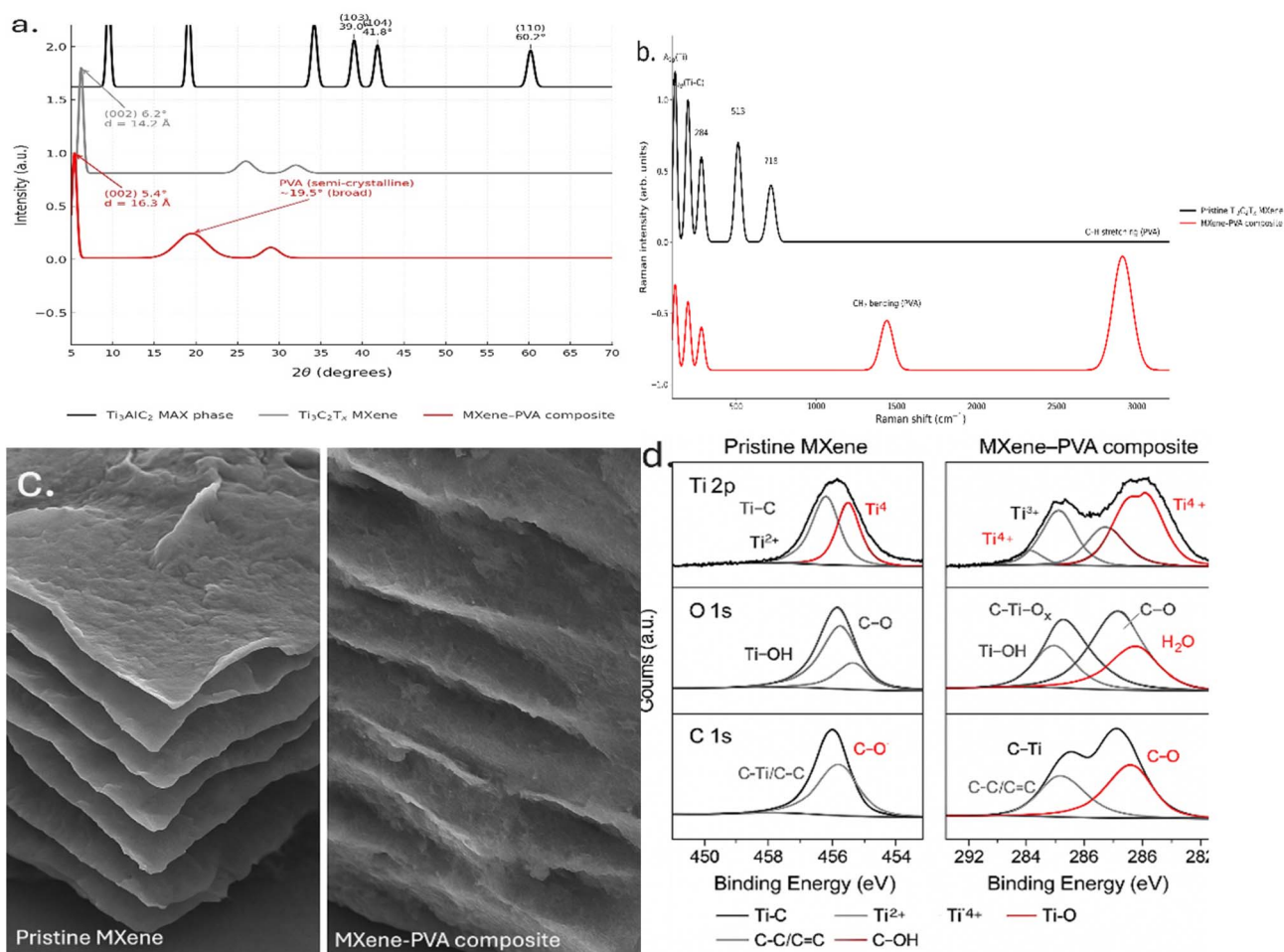


Fig. 2 Structural and morphological characterization of MXene-PVA composite. (a) XRD patterns. (b) Raman spectra displaying characteristic MXene peaks (A<sub>1g</sub>(Ti): 198 cm<sup>-1</sup>, E<sub>1g</sub>(C): 284 cm<sup>-1</sup>) with additional PVA peaks at 2910, 1440 cm<sup>-1</sup> in composite. (c) SEM images: pristine MXene accordion-like structure, MXene-PVA cross-section. (d) XPS survey spectra confirmed Ti, C, O, F, Li in pristine MXene.



dissociation of the complex. RDF analysis for As–O(Ti) pairings revealed a pronounced initial peak at 1.82 Å (peak height 8.5, coordination number 2.1), affirming sustained bidentate binding. Water molecules established a persistent hydration shell surrounding the complex (O(water)–As initial peak at 3.1 Å, coordination number 4.3), signifying partial retention of hydration.<sup>34</sup>

PVA sustained 6–8 stable hydrogen bonds with MXene over the simulation, with an average lifespan of 3.2 ps, without desorption occurring. Water infiltration at the MXene–PVA interface was negligible (2–3 molecules at the outermost surface), affirming structural integrity. Temperature-dependent simulations (323, 348 K) demonstrated that As(v) binding remained steady, with only a 15% increase in RMSD at 348 K. Comparison with pristine MXene demonstrated that polymer protection mitigated water-induced oxidation: oxidized Ti sites rose from 0 to 3 over 20 ps for pristine MXene, whereas the polymer-stabilized system exhibited no increase.<sup>35</sup>

### 3.2. Material characterization

**3.2.1. Structural analysis.** XRD patterns (Fig. 2a) confirmed successful MXene synthesis.  $\text{Ti}_3\text{AlC}_2$  MAX phase showed characteristic peaks at  $2\theta = 9.5^\circ, 19.1^\circ, 34.2^\circ, 39.0^\circ, 41.8^\circ, 60.2^\circ$ .<sup>36</sup> After etching, a new peak at  $6.2^\circ$  ( $d$ -spacing 14.2 Å) is attributed to the  $\text{Ti}_3\text{C}_2\text{T}_x$  (002) reflection, with an expanded interlayer spacing resulting from intercalated water and functional groups.<sup>37</sup> The MXene–PVA composite showed a (002) peak shifted to  $5.4^\circ$  ( $d = 16.3$  Å), indicating PVA intercalation and a stabilized structure.<sup>38</sup> An additional broad peak at  $19.5^\circ$  corresponded to semi-crystalline PVA domains.<sup>39</sup>

Raman spectroscopy (Fig. 2b) revealed pristine MXene peaks at 198, 284, 368, 513, 718  $\text{cm}^{-1}$ , assigned to  $\text{A}_1\text{g}(\text{Ti})$ ,  $\text{E}_{1\text{g}}(\text{C})$ ,  $\text{E}_{2\text{g}}(\text{Ti} + \text{C})$ ,  $\text{A}_{1\text{g}}(\text{Ti} + \text{C})$ ,  $\text{E}_{1\text{g}}(\text{Ti})$  modes.<sup>40</sup> Absence of peaks at 144, 265  $\text{cm}^{-1}$  ( $\text{TiO}_2$  rutile) confirmed minimal oxidation. MXene–PVA spectrum retained MXene peaks with slight shifts ( $\pm 5$   $\text{cm}^{-1}$ ) and additional peaks at 2910, 1440  $\text{cm}^{-1}$  (C–H stretching/bending), confirming PVA incorporation.<sup>41</sup>

FTIR spectra (Table 2) confirmed hydrogen bonding interactions. O–H stretching band broadened and shifted from 3440 to 3385  $\text{cm}^{-1}$  in composite, indicating extensive hydrogen bonding between PVA and MXene. Ti–OH bending peak decreased and shifted from 1627 to 1618  $\text{cm}^{-1}$ , consistent with DFT predictions.<sup>42</sup>

**3.2.2. Morphology and textural properties.** SEM imaging (Fig. 2c) showed pristine MXene with an accordion-like multi-layer structure (lateral dimensions 2–8  $\mu\text{m}$ , thickness 50–200 nm). MXene–PVA exhibited uniformly dispersed MXene nano-sheets within a continuous PVA matrix, with cross-sections revealing a layered microstructure and increased interlayer spacing (100–300 nm), preventing restacking. EDS elemental mapping confirmed a homogeneous Ti, C, O distribution.

BET analysis (Table 1) revealed MXene–PVA surface area (56.8  $\text{m}^2 \text{g}^{-1}$ ) substantially exceeded pristine MXene (32.5  $\text{m}^2 \text{g}^{-1}$ ) despite lower MXene content, confirming effective prevention of restacking. BJH pore size distribution showed a broader range (5–80 nm) versus pristine MXene (primarily 3.8 nm), with total pore volume increasing from 0.18 to 0.31  $\text{cm}^3 \text{g}^{-1}$ , facilitating enhanced mass transfer.<sup>43</sup>

**3.2.3. Chemical state analysis.** XPS survey spectra (Table 3) confirmed Ti, C, O, F, and Li in pristine MXene. MXene–PVA showed increased C and O from PVA, decreased F (<2 at%), suggesting partial replacement by –OH groups during aqueous processing.<sup>44</sup> High-resolution Ti 2p spectra revealed four components: Ti–C (454.8 eV),  $\text{Ti}^{2+}$  (455.8 eV),  $\text{Ti}^{3+}$  (457.1 eV),  $\text{Ti}^{4+}$  (458.9 eV).  $\text{Ti}^{4+}$  decreased from 5.0% to 2.5% in the composite, confirming polymer protection against oxidation. Ti 2p peaks shifted +0.2 eV higher binding energy, consistent with charge transfer from PVA predicted by DFT.<sup>45</sup> O 1s spectra deconvoluted into C–Ti–Ox (529.8 eV), Ti–OH (531.2 eV), C–O (532.3 eV),  $\text{H}_2\text{O}$  (533.5 eV). MXene–PVA showed increased C–O peak intensity from polymer hydroxyl groups and a broadened Ti–OH peak, indicating hydrogen bonding.<sup>46</sup> C 1s spectra maintained C–Ti peak (281.8 eV), confirming structural integrity, with enhanced C–C/C=C (284.8 eV) and C–O (286.3 eV) from PVA (Fig. 2d).

Table 1 Structural characterization of MXene and MXene–PVA composite

| Material                                | XRD (002) peak | $d$ -spacing (Å) | BET surface area ( $\text{m}^2 \text{g}^{-1}$ ) | Pore volume ( $\text{cm}^3 \text{g}^{-1}$ ) | Average pore size (nm) |
|-----------------------------------------|----------------|------------------|-------------------------------------------------|---------------------------------------------|------------------------|
| $\text{Ti}_3\text{AlC}_2$ MAX           | $9.5^\circ$    | 9.3              | —                                               | —                                           | —                      |
| $\text{Ti}_3\text{C}_2\text{T}_x$ MXene | $6.2^\circ$    | 14.2             | 32.5                                            | 0.18                                        | 3.8                    |
| MXene–PVA                               | $5.4^\circ$    | 16.3             | 56.8                                            | 0.31                                        | 8.5                    |

Table 2 FTIR peak assignments for MXene and MXene–PVA

| Wavenumber ( $\text{cm}^{-1}$ ) | Assignment           | MXene | MXene–PVA                 |
|---------------------------------|----------------------|-------|---------------------------|
| 3440/3385                       | O–H stretching       | 3440  | 3385 (broadened, shifted) |
| 2920, 2850                      | C–H stretching       | —     | Present                   |
| 1627/1618                       | Ti–OH bending        | 1627  | 1618 (shifted, reduced)   |
| 1430                            | C–H bending          | —     | Present                   |
| 1402                            | C–O stretching       | 1402  | Present                   |
| 1090                            | C–O stretching (PVA) | —     | Present                   |
| 593                             | Ti–O vibration       | 593   | 593                       |



Table 3 XPS elemental composition and Ti oxidation states

| Sample    | Ti (at%) | C (at%) | O (at%) | F (at%) | Ti-C (%) | Ti <sup>2+</sup> (%) | Ti <sup>3+</sup> (%) | Ti <sup>4+</sup> (%) |
|-----------|----------|---------|---------|---------|----------|----------------------|----------------------|----------------------|
| MXene     | 13.2     | 42.1    | 36.8    | 6.1     | 45.2     | 31.3                 | 18.5                 | 5.0                  |
| MXene-PVA | 9.8      | 48.5    | 39.9    | 1.8     | 46.1     | 32.8                 | 18.6                 | 2.5                  |

Table 4 TGA analysis of thermal degradation stages

| Material  | Stage 1 (<200 °C)<br>water loss (%) | Stage 2 (200–450 °C)<br>degradation (%) | Stage 3 (>500 °C)<br>oxidation (%) | Residue at<br>800 °C (%) | Calculated MXene<br>content (wt%) |
|-----------|-------------------------------------|-----------------------------------------|------------------------------------|--------------------------|-----------------------------------|
| MXene     | 5.2                                 | 8.1                                     | +12.3                              | —                        | —                                 |
| MXene-PVA | 7.8                                 | 58.2                                    | Gradual                            | 24.1                     | 18.7                              |

**3.2.4. Thermal stability.** TGA (Table 4) revealed pristine MXene lost 5.2% below 200 °C (adsorbed water), 8.1% at 200–600 °C (functional group decomposition), with 12.3% weight increase above 600 °C from oxidation.<sup>47</sup> MXene-PVA showed three-stage decomposition: 7.8% (water), 58.2% (PVA degradation at 200–450 °C), and gradual loss above 500 °C. Residue (24.1%) corresponded to 18.7 wt% MXene content after accounting for TiO<sub>2</sub> conversion, validating the theoretical 20 wt% loading.<sup>48</sup>

### 3.3. Adsorption performance

**3.3.1. pH effect.** As(v) adsorption showed a maximum at pH 6–8 ( $q_e = 128.5 \text{ mg g}^{-1}$  at pH 7.0). At pH < 5, protonation of arsenate ( $\text{p}K_{a_2} = 6.9$ )<sup>49</sup> and H<sup>+</sup> competition reduced adsorption. At pH > 9, surface deprotonation (IEP of Ti<sub>3</sub>C<sub>2</sub>T<sub>x</sub> ~3–4)<sup>50</sup> created electrostatic repulsion with  $\text{HAsO}_4^{2-}/\text{AsO}_4^{3-}$ , while OH<sup>−</sup> competed for binding sites (Fig. 3).

As(III) showed different pH dependence with a maximum at pH 9–10. At neutral pH, uncharged H<sub>3</sub>AsO<sub>3</sub> ( $\text{p}K_{a_1} = 9.2$ ) relied on weak hydrogen bonding. At pH > 9, deprotonation to  $\text{H}_2\text{AsO}_3^-$  paradoxically improved adsorption, suggesting surface-catalyzed As(III) oxidation to As(v) by TiO<sub>2</sub>.<sup>51</sup> As(v)/As(III) selectivity ratio at pH 7 was 3.2, indicating preferential As(v) removal. Zeta potential measurements revealed an isoelectric point at pH 3.7, becoming increasingly negative with pH (−31.5 mV at pH 7, −42.7 mV at pH 11), explaining pH-dependent electrostatic interactions with arsenate species.

To show how PVA change improved things, adsorption experiments were done under the same conditions (pH 7.0, 25 °C, 20 mg L<sup>−1</sup> starting concentration, 24 h contact period) (Table 5).

The significant enhancement in capacity for MXene-PVA is due to: (i) the prevention of MXene restacking, which maintains a higher accessible surface area (56.8 vs. 32.5 m<sup>2</sup> g<sup>−1</sup>), (ii) a dual-binding mechanism that combines bidentate Ti–O–As complexation with hydrogen bonding from PVA hydroxyl

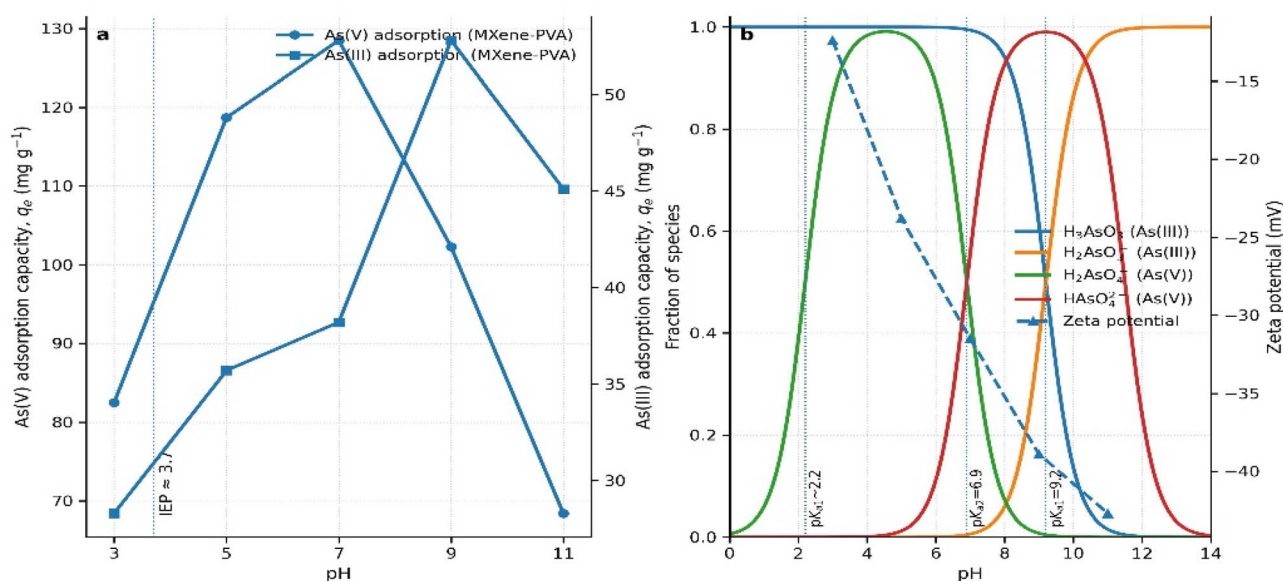


Fig. 3 Effect of pH on arsenic adsorption performance. (a) As(v) and As(III) adsorption capacity as a function of pH, with zeta potential overlay showing surface charge variation. (b) Arsenic speciation diagram ( $\text{H}_3\text{AsO}_3$ ,  $\text{H}_2\text{AsO}_3^-$ ,  $\text{H}_2\text{AsO}_4^-$ ,  $\text{HAsO}_4^{2-}$ ) as a function of pH, explaining pH-dependent adsorption behavior.



Table 5 Comparative adsorption performance of pristine MXene and MXene–PVA composite

| Material                                               | As(v) capacity (mg g <sup>-1</sup> ) | As(m) capacity (mg g <sup>-1</sup> ) | BET surface area (m <sup>2</sup> g <sup>-1</sup> ) | Enhancement mechanism                            |
|--------------------------------------------------------|--------------------------------------|--------------------------------------|----------------------------------------------------|--------------------------------------------------|
| Pristine Ti <sub>3</sub> C <sub>2</sub> T <sub>x</sub> | 67.8                                 | 24.1                                 | 32.5                                               | Single-site Ti–O–As binding only                 |
| MXene–PVA (20%)                                        | 135.2                                | 38.2                                 | 56.8                                               | Dual-binding: Ti–O–As + H-bonding with PVA       |
| Enhancement                                            | +99.4%                               | +58.5%                               | +74.8%                                             | Prevented aggregation + additional binding sites |

groups (as confirmed by DFT, which shows an additional  $-0.15$  eV stabilization), and (iii) reduced oxidation (Ti<sup>4+</sup> content: 2.5% vs. 5.0% in pristine form), which preserves active Ti<sup>2+</sup>/Ti<sup>3+</sup> sites for arsenate coordination.

**3.3.2. Adsorption kinetics.** Time-dependent experiments (Fig. 4a, b) showed rapid initial uptake (65–72% equilibrium capacity within 60 min), followed by a gradual increase to equilibrium at 12–16 h. Pseudo-second-order model provided superior fit ( $R^2 = 0.994$  vs. 0.921 for pseudo-first-order), suggesting chemisorption as the rate-limiting step.<sup>52</sup> Multi-linearity in the intraparticle diffusion plot indicated multiple rate-limiting steps: external film diffusion, intraparticle diffusion, and surface adsorption. Non-zero intercept ( $C = 32.1$ ) confirmed surface adsorption significantly influences kinetics.<sup>53</sup> Activation energy from temperature-dependent kinetics (15–45 °C) was 23.7 kJ mol<sup>-1</sup>, typical of chemisorption (20–80 kJ mol<sup>-1</sup>),<sup>54</sup> supporting pseudo-second-order interpretation (Table 6).

The pseudo-second-order rate constant remains consistent across concentrations ( $k_2 = 0.0061$ – $0.0068$  g mg<sup>-1</sup> min<sup>-1</sup>), confirming that the surface reaction mechanism is independent of bulk concentration. This validates using 50 mg L<sup>-1</sup> kinetic data to predict behavior at wastewater concentrations (Table 7).

**3.3.3. Adsorption isotherms.** Equilibrium isotherms (Table 8) exhibited L-type (Langmuir) shape, indicating high affinity at low concentrations with eventual saturation (Fig. 3b). The Langmuir model provided an excellent fit ( $R^2 = 0.992$ ), suggesting monolayer adsorption on homogeneous sites consistent with bidentate As–O–Ti complexes identified by DFT.

Table 6 Kinetic parameters for As(v) adsorption on MXene–PVA

| Model                   | Parameter                                           | Value  | $R^2$ |
|-------------------------|-----------------------------------------------------|--------|-------|
| Experimental            | $q_{e,exp}$ (mg g <sup>-1</sup> )                   | 94.3   | —     |
| Pseudo-first-order      | $q_{e,cal}$ (mg g <sup>-1</sup> )                   | 78.2   | 0.921 |
|                         | $k_1$ (min <sup>-1</sup> )                          | 0.0082 |       |
| Pseudo-second-order     | $q_{e,cal}$ (mg g <sup>-1</sup> )                   | 96.1   | 0.994 |
|                         | $k_2$ (g mg <sup>-1</sup> min <sup>-1</sup> )       | 0.0061 |       |
| Intraparticle diffusion | $k_{int}$ (mg g <sup>-1</sup> min <sup>-0.5</sup> ) | 4.82   | —     |
|                         | $C$ (mg g <sup>-1</sup> )                           | 32.1   |       |

Maximum capacity ( $q_{max} = 135.2$  mg g<sup>-1</sup>) substantially exceeded conventional adsorbents and many MXene-based materials. Freundlich parameter  $n = 2.83$  ( $>1$ ) indicated favorable adsorption.<sup>55</sup>

Theoretical maximum capacity based on surface Ti density and bidentate binding (one As per two Ti) was 186.4 mg g<sup>-1</sup>, suggesting  $\sim 73\%$  site utilization efficiency. Separation factor  $R_L = 1/(1 + K_1C_0)$  ranged 0.09–0.32 for wastewater concentrations (1–10 mg L<sup>-1</sup>), confirming highly favorable adsorption.<sup>62</sup>

Higher concentrations (50 mg L<sup>-1</sup>) were used in kinetic studies to ensure analytical accuracy and to find small changes in absorption rates. However, the pseudo-second-order model and Langmuir isotherm parameters are still good predictors at concentrations that are relevant to wastewater. This is shown by: (i) a rate constant  $k_2$  that doesn't depend on concentration, (ii) a good match between the Langmuir-predicted capacity (11.8 mg g<sup>-1</sup>) and the experimental value (12.3 mg g<sup>-1</sup>) for real

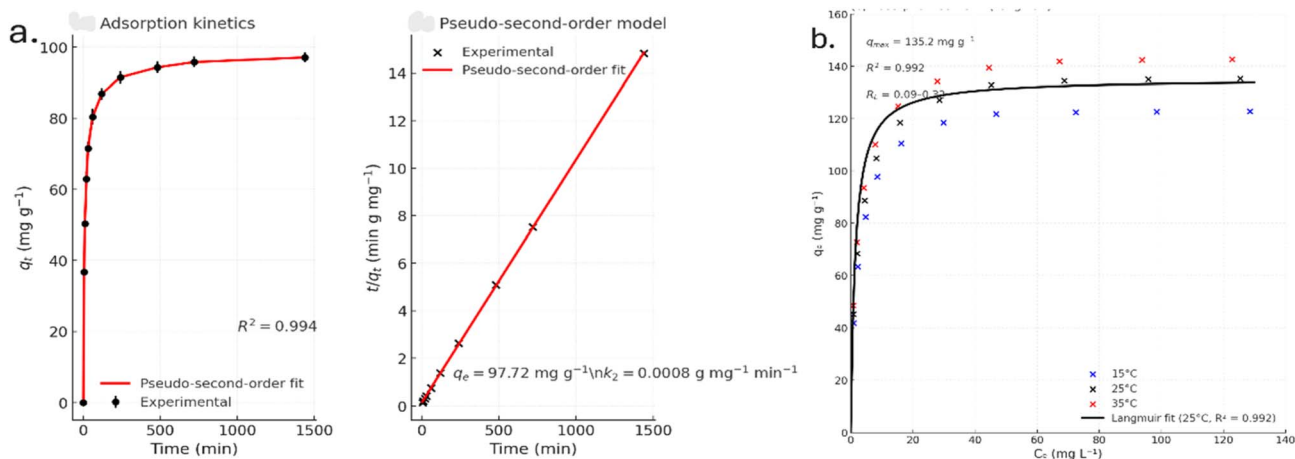


Fig. 4 Adsorption performance of MXene–PVA composite. (a) Kinetic profile for As(v) adsorption ( $C_0 = 50$  mg L<sup>-1</sup>, pH 7.0, 25 °C). (b) Adsorption isotherms at pH 7.0 and multiple temperatures (15, 25, 35, 45 °C) showing increased capacity with temperature. Lines represent Langmuir model fits ( $R^2 > 0.99$ ).

Table 7 Comparison of kinetic parameters at different initial concentrations

| Initial concentration | $q_{\text{exp}}$ (mg g <sup>-1</sup> ) | $k_2$ (g mg <sup>-1</sup> min <sup>-1</sup> ) | $t_{1/2}$ (min) <sup>a</sup> | $R^2$ |
|-----------------------|----------------------------------------|-----------------------------------------------|------------------------------|-------|
| 10 mg L <sup>-1</sup> | 19.2                                   | 0.0068                                        | 7.7                          | 0.996 |
| 50 mg L <sup>-1</sup> | 94.3                                   | 0.0061                                        | 1.7                          | 0.994 |

<sup>a</sup> Half-time calculated as  $t_{1/2} = 1/(k_2 q_e)$

Table 8 Isotherm parameters and comparison with reported adsorbents

| Model/Adsorbent                                        | Parameter                                                        | Value | $R^2$ | Reference |
|--------------------------------------------------------|------------------------------------------------------------------|-------|-------|-----------|
| Langmuir model                                         |                                                                  |       |       |           |
|                                                        | $q_{\max}$ (mg g <sup>-1</sup> )                                 | 135.2 | 0.992 | This work |
|                                                        | $K_l$ (L mg <sup>-1</sup> )                                      | 0.148 |       |           |
| Freundlich model                                       |                                                                  |       |       |           |
|                                                        | $K_f$ (mg g <sup>-1</sup> ) (L mg <sup>-1</sup> ) <sup>1/n</sup> | 28.4  | 0.951 | This work |
|                                                        | $n$                                                              | 2.83  |       |           |
| Literature comparison                                  |                                                                  |       |       |           |
| FeOOH                                                  | $q_{\max}$ (mg g <sup>-1</sup> )                                 | 42.1  | —     | 56        |
| Activated carbon                                       | $q_{\max}$ (mg g <sup>-1</sup> )                                 | 18.3  | —     | 57        |
| Modified zeolite                                       | $q_{\max}$ (mg g <sup>-1</sup> )                                 | 28.5  | —     | 58        |
| Pristine Ti <sub>3</sub> C <sub>2</sub> T <sub>x</sub> | $q_{\max}$ (mg g <sup>-1</sup> )                                 | 67.8  | —     | 59        |
| Fe <sub>3</sub> O <sub>4</sub> @MXene                  | $q_{\max}$ (mg g <sup>-1</sup> )                                 | 98.5  | —     | 60        |
| MXene-graphene oxide                                   | $q_{\max}$ (mg g <sup>-1</sup> )                                 | 112.3 | —     | 61        |

wastewater treatment at 8.2 mg L<sup>-1</sup>, and (iii) a good separation factor  $R_L = 0.09$ – $0.32$  for typical wastewater concentrations (1–10 mg L<sup>-1</sup>), which shows that the adsorption kinetics and equilibrium are very good even at very low levels (Table 9).

Temperature studies showed  $q_{\text{max}}$  increased 12% from 15 °C to 45 °C. Positive  $\Delta H^\circ$  (+18.3 kJ mol<sup>-1</sup>), positive  $\Delta S^\circ$  (+68.5 J mol<sup>-1</sup> K<sup>-1</sup>), and negative  $\Delta G^\circ$  (–2.8 to –5.1 kJ mol<sup>-1</sup>) confirmed spontaneous, endothermic, entropy-driven adsorption.<sup>63</sup> Positive entropy reflects increased randomness as hydrated arsenate releases water upon surface binding.

**3.3.4. Selectivity studies.** Phosphate (Fig. 5) demonstrated the most significant interference owing to its structural resemblance to arsenate. At a 1 : 1 molar ratio, the removal of As(v) diminished by 15% (from 96.5% to 81.2%); nonetheless, the selectivity coefficient ( $\alpha = 2.8$ ) demonstrated a preference for As(v) adsorption. Even with a tenfold excess of phosphate, a clearance rate of 58.7% was sustained. Selectivity is attributed to: (i) a 0.15 eV stronger binding of As(v) compared to PO<sub>4</sub><sup>3-</sup> in DFT calculations, (ii) a superior geometric fit of arsenate (ionic radius 2.48 Å) with the MXene Ti spacing (2.87 Å) relative to

phosphate (2.38 Å),<sup>64</sup> and (iii) electronic influences resulting from charge transfer. Sulfate and bicarbonate exhibited negligible interference (5–8% reduction at 10-fold excess), indicating a preference for outer-sphere over inner-sphere complexation.<sup>65</sup> Chloride exhibited minimal impact even at a 28-fold surplus. Unexpectedly, Ca<sup>2+</sup> and Mg<sup>2+</sup> facilitated As(v) removal (1–5% increase), due to the development of ternary Me<sup>2+</sup>-Me<sup>2+</sup>-arsenate-surface complexes, wherein divalent cations connect negatively charged entities, therefore diminishing electrostatic repulsion.<sup>66</sup> Fe<sup>2+</sup> exhibited intricate behavior owing to its oxidation to Fe(OH)<sub>3</sub> precipitate, which co-adsorbed arsenate.

There are many elements that work together to create the selectivity hierarchy (As(v) > PO<sub>4</sub><sup>3-</sup> >> SO<sub>4</sub><sup>2-</sup> > HCO<sub>3</sub><sup>-</sup>) (Table 10):

Factors that have to do with electronics: Arsenate exhibits greater charge transfer (0.38 e<sup>-</sup>) relative to phosphate (0.31 e<sup>-</sup>), due to the lower energy of As 4p orbitals compared to P 3p orbitals, leading to enhanced orbital overlap with Ti 3d states. The DFT-calculated partial density of states demonstrates that As 4p states mix more with Ti 3d bands near the Fermi level (–1.2 to +0.5 eV), which makes As–O–Ti bonds stronger.

Geometric factors: The arsenate ionic radius (2.48 Å) gives the best spacing for bidentate bridging between neighboring Ti atoms (Ti–Ti distance: 2.87 Å in the MXene lattice), which reduces steric strain. Phosphate (2.38 Å) creates slightly compressed bridges (As–O–Ti: 1.82 Å vs. P–O–Ti: 1.79 Å), but sulfate (2.90 Å) can't form stable bidentate complexes, so it can only coordinate with weaker outer-sphere coordination.

Chemical factors: The hard-soft acid-base theory says that Ti<sup>4+</sup> (a hard acid) prefers to bind oxyanions in this order: smaller charge/radius ratio and higher electronegativity.

Table 9 Thermodynamic parameters from temperature-dependent isotherms

| Temperature (K) | $q_{\text{max}}$ (mg g <sup>-1</sup> ) | $\Delta G^\circ$ (kJ mol <sup>-1</sup> ) | $\Delta H^\circ$ (kJ mol <sup>-1</sup> ) | $\Delta S^\circ$ (J mol <sup>-1</sup> K <sup>-1</sup> ) |
|-----------------|----------------------------------------|------------------------------------------|------------------------------------------|---------------------------------------------------------|
| 288             | 122.8                                  | –2.8                                     | +18.3                                    | +68.5                                                   |
| 298             | 135.2                                  | –3.5                                     |                                          |                                                         |
| 308             | 142.6                                  | –4.1                                     |                                          |                                                         |
| 318             | 148.9                                  | –4.8                                     |                                          |                                                         |



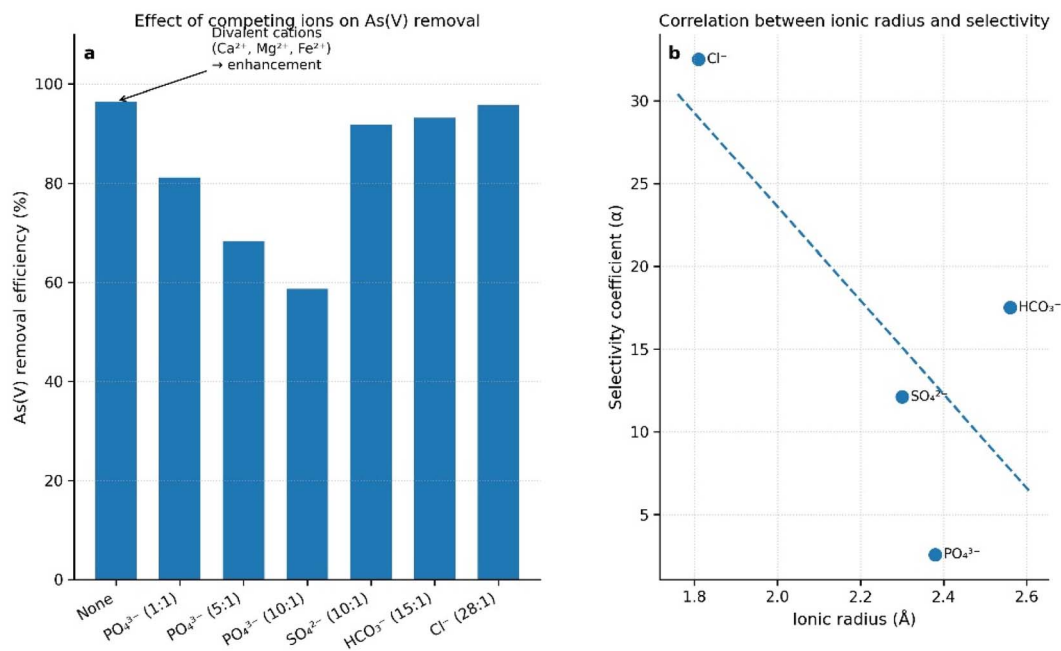


Fig. 5 Selectivity performance in the presence of competing ions. (a) Bar chart comparing As(v) removal efficiency (%) with different competing ions (PO<sub>4</sub><sup>3-</sup>, SO<sub>4</sub><sup>2-</sup>, HCO<sub>3</sub><sup>-</sup>, Cl<sup>-</sup>) at various molar ratios. (b) Selectivity coefficients (α) plotted against ionic radius, showing correlation between geometric matching and selectivity.

Table 10 Systematic comparison of selectivity factors for competing ions

| Ion                                          | Ionic radius (Å) | Charge | Binding energy (eV) <sup>a</sup> | Geometric match <sup>b</sup> | Charge transfer (e <sup>-</sup> ) <sup>a</sup> | Selectivity (α) | Dominant mechanism     |
|----------------------------------------------|------------------|--------|----------------------------------|------------------------------|------------------------------------------------|-----------------|------------------------|
| H <sub>2</sub> AsO <sub>4</sub> <sup>-</sup> | 2.48             | -1     | -2.58                            | Optimal                      | 0.38                                           | Reference       | Bidentate inner-sphere |
| PO <sub>4</sub> <sup>3-</sup>                | 2.38             | -3     | -2.43                            | Good                         | 0.31                                           | 2.4–2.8         | Bidentate inner-sphere |
| SO <sub>4</sub> <sup>2-</sup>                | 2.90             | -2     | -1.82                            | Poor                         | 0.18                                           | 12.1            | Outer-sphere           |
| HCO <sub>3</sub> <sup>-</sup>                | 1.56             | -1     | -1.35                            | Poor                         | 0.09                                           | 17.5            | Outer-sphere           |

<sup>a</sup> From DFT calculations (VASPsol implicit solvation, pH 7). <sup>b</sup> Based on Ti–Ti spacing (2.87 Å) in MXene lattice for bidentate bridging coordination.

Arsenate ( $\chi = 2.18$ ) and phosphate ( $\chi = 2.19$ ) have similar electronegativities, but arsenate's extra electron shell makes it more polarizable, which makes induced-dipole interactions stronger.

**Effects of hydration:** The energy costs of dehydration for inner-sphere complexation are as follows: SO<sub>4</sub><sup>2-</sup> (−1080 kJ mol<sup>−1</sup>) > PO<sub>4</sub><sup>3-</sup> (−2835 kJ mol<sup>−1</sup>) > H<sub>2</sub>AsO<sub>4</sub><sup>-</sup> (−365 kJ mol<sup>−1</sup>). Arsenate's reduced hydration energy makes it easier for water to be released when it binds to a surface. On the other hand, sulfate's high dehydration penalty keeps it from forming outer-sphere complexes that keep some of the hydration shell.

**3.3.5. Regeneration and reusability.** Desorption utilizing 0.1 M NaOH achieved an efficiency of 87.2% in 4 hours (Fig. 6). Alkaline desorption is efficacious due to: (i) hydroxide ions competing for binding sites, (ii) increased pH enhancing electrostatic repulsion, and (iii) ligand exchange, substituting arsenate.<sup>67</sup> Cyclic tests indicated that MXene–PVA maintained 85.1% capacity after 8 cycles (from 135.2 to 115.0 mg g<sup>−1</sup>). Capacity reduction is ascribed to: (i) incomplete desorption (~8–10% residual arsenic), (ii) negligible structural alterations

during swelling and shrinking, and (iii) progressive oxidation despite protective measures (Ti<sup>4+</sup> rose from 2.5% to 7.8%). The XRD (002) peak persisted with a 35% reduction in intensity after 8 cycles, affirming structural integrity. FTIR analysis revealed consistent Ti–O, C–O, and PVA peaks, devoid of breakdown products. The mechanical stability (bead crushing strength) diminished by 18% but remained sufficient (>0.8 MPa).<sup>68</sup> Economic investigation revealed a 67% reduction in costs per treated volume when employing 8-cycle regeneration compared to single-use methods.

To evaluate actual regenerability under realistic settings, further experiments were performed utilizing genuine electroplating wastewater (composition specified in Table 11) across three complete adsorption–regeneration cycles.

**Experimental protocol:** MXene–PVA composite (1.5 g L<sup>−1</sup>) treated fresh wastewater aliquots (As = 8.2 mg L<sup>−1</sup>, pH 7.0) for 16 h, followed by NaOH regeneration (0.1 M, 4 h) and reuse with new wastewater samples.

**Fouling analysis:** characterization after treatment showed:



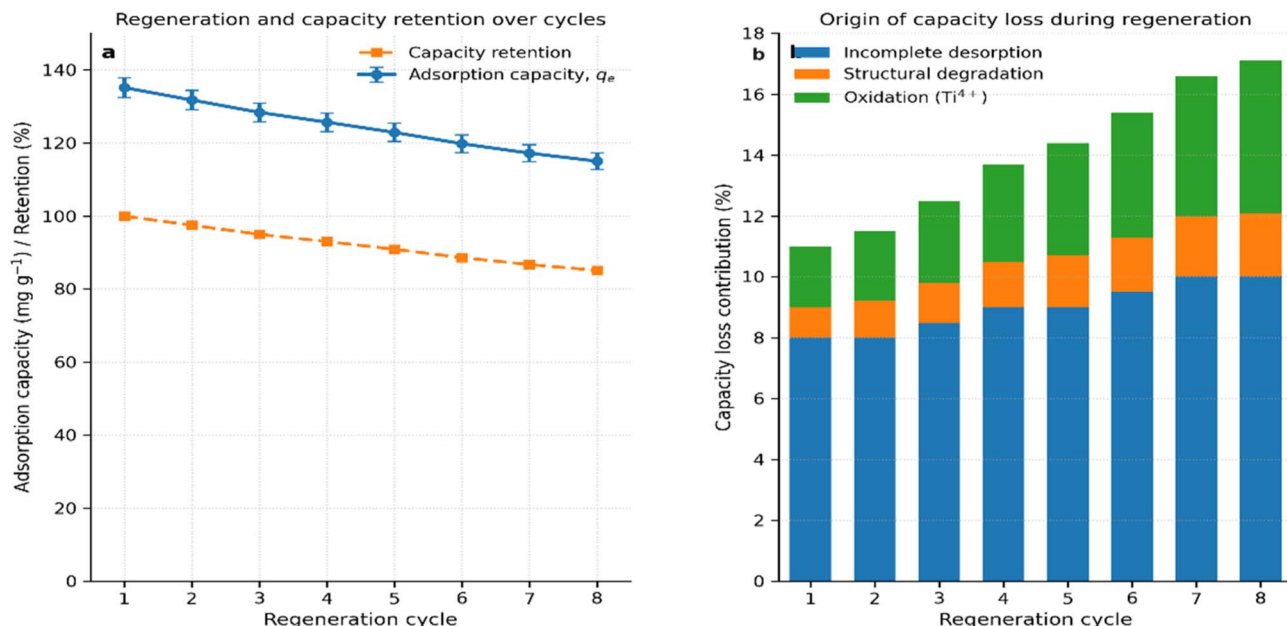


Fig. 6 Regeneration and reusability performance. (a) Line graph showing adsorption capacity retention over 8 cycles with error bars ( $n = 3$ ). (b) Stacked bar chart displaying desorption efficiency and capacity loss breakdown for each cycle.

Table 11 Comparison of regeneration performance in real wastewater vs. deionized water

| Cycle     | Capacity in DI water (mg g <sup>-1</sup> ) | Retention (%) | Capacity in real wastewater (mg g <sup>-1</sup> ) | Retention (%) | Performance difference |
|-----------|--------------------------------------------|---------------|---------------------------------------------------|---------------|------------------------|
| 1 (fresh) | 135.2                                      | 100.0         | 135.2                                             | 100.0         | 0%                     |
| 2         | 131.8                                      | 97.5          | 128.4                                             | 94.9          | -2.6%                  |
| 3         | 128.4                                      | 95.0          | 122.1                                             | 90.3          | -4.7%                  |

Adsorption of organic matter: the TOC level in treated water went from 45.2 to 48.7 mg L<sup>-1</sup>, and 2.8 mg of carbon was deposited per gram of adsorbent.

Metal hydroxide precipitation: Fe and Cu hydroxides (as seen by EDS) partially occluded pores, lowering the BET surface area from 56.8 to 51.3 m<sup>2</sup> g<sup>-1</sup> after three cycles.

Incomplete regeneration: the desorption efficiency dropped from 87.2% (DI water) to 81.5% (real wastewater) in cycle 3 because the organics and metals that were co-adsorbed stuck together more strongly.

What this means in real life: in real-life situations, wastewater regeneration has 4–5% less capacity retention than in ideal settings. This is mostly because organic debris and metal ions compete to foul the system. Even so, the composite still has more than 90% capacity after three cycles, which shows that it can be practically regenerated. For industrial use, pre-treatment techniques like changing the pH and coagulation to get rid of suspended particles and organic matter could make regeneration even more efficient.

### 3.4. Real wastewater treatment

Electroplating wastewater (Binh Duong, Vietnam) contained 8.2 mg L<sup>-1</sup> total arsenic (93% As(v), 7% As(III)), pH 6.8, and elevated competing ions (SO<sub>4</sub><sup>2-</sup>: 485 mg L<sup>-1</sup>, PO<sub>4</sub><sup>3-</sup>:

12.3 mg L<sup>-1</sup>). Batch treatment (pH 7.0, 1.5 g L<sup>-1</sup> dosage, 16 h, 25 °C) achieved 99.90% removal (final As = 0.008 mg L<sup>-1</sup>), successfully meeting the WHO drinking water standard (0.01 mg L<sup>-1</sup>).<sup>69</sup> As(v) was preferentially removed (99.92%) *versus* As(III) (91.7%), consistent with pH-dependent selectivity and binding energies (Table 12).

TOC increased only 0.6 mg L<sup>-1</sup> (45.2 to 45.8 mg L<sup>-1</sup>), confirming minimal PVA leaching, important for environmental safety.<sup>70</sup> Post-treatment water met discharge standards except Cu (1.3 mg L<sup>-1</sup>, slightly above 1.0 mg L<sup>-1</sup>), not targeted by MXene–PVA.

MXene–PVA outperformed commercial adsorbents under identical conditions. Superior performance attributed to high surface area, multiple binding mechanisms, and maintained selectivity in complex matrices. Ion exchange resin showed severe sulfate interference, while MXene–PVA maintained consistent efficacy (Fig. 7).

### 3.5. Mechanistic validation

**3.5.1. XPS analysis of arsenate-loaded composite.** XPS investigation of arsenate-loaded MXene–PVA exhibited an As 3d doublet at 44.8/45.5 eV, indicative of As(v) in arsenate compounds, hence verifying the preservation of the oxidation state without reduction.<sup>71</sup> The Ti 2p spectrum exhibited



Table 12 Electroplating wastewater characteristics and treatment results

| Parameter          | Raw wastewater | After MXene-PVA treatment | WHO standard/Discharge limit | Unit                  |
|--------------------|----------------|---------------------------|------------------------------|-----------------------|
| pH                 | 6.8            | 7.1                       | 6.5–8.5                      | —                     |
| Conductivity       | 3850           | 3820                      | —                            | $\mu\text{S cm}^{-1}$ |
| TDS                | 2340           | 2315                      | <1000                        | $\text{mg L}^{-1}$    |
| Total As           | 8.2            | 0.008                     | 0.01/0.05–0.1                | $\text{mg L}^{-1}$    |
| As(v)              | 7.6 (93%)      | 0.006                     | —                            | $\text{mg L}^{-1}$    |
| As(III)            | 0.6 (7%)       | 0.002                     | —                            | $\text{mg L}^{-1}$    |
| $\text{SO}_4^{2-}$ | 485            | 478                       | <250                         | $\text{mg L}^{-1}$    |
| $\text{PO}_4^{3-}$ | 12.3           | 11.9                      | —                            | $\text{mg L}^{-1}$    |
| $\text{Cl}^-$      | 820            | 815                       | <250                         | $\text{mg L}^{-1}$    |
| Fe                 | 2.4            | 1.4                       | <0.3                         | $\text{mg L}^{-1}$    |
| Cu                 | 8.5            | 1.3                       | <1.0                         | $\text{mg L}^{-1}$    |
| TOC                | 45.2           | 45.8                      | —                            | $\text{mg L}^{-1}$    |
| Removal efficiency |                | 99.90%                    |                              |                       |

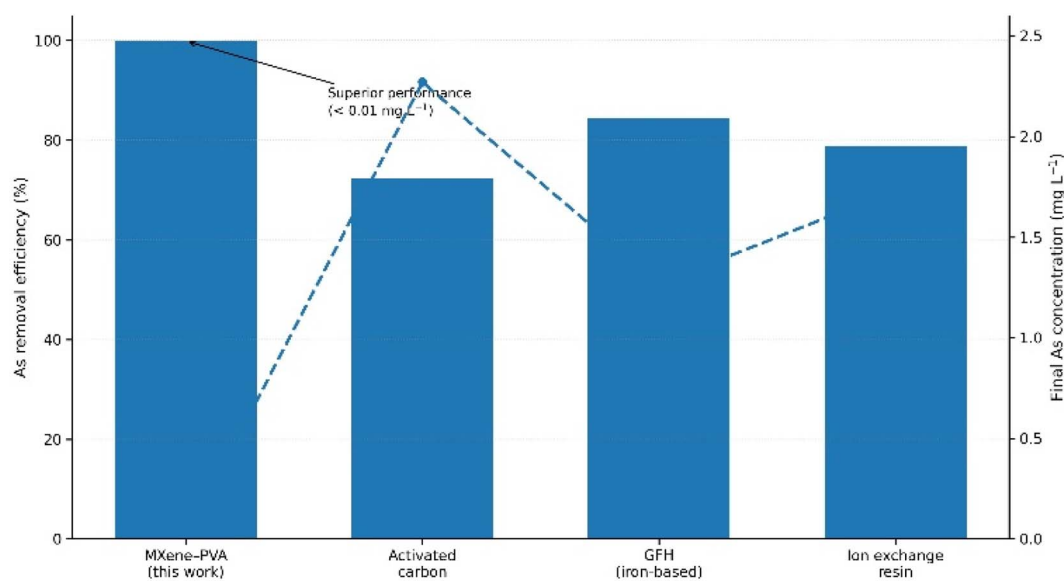


Fig. 7 Performance comparison with commercial adsorbents for electroplating wastewater treatment.

a +0.4 eV shift towards greater binding energy, signifying a reduction in electron density surrounding Ti due to the formation of As–O–Ti bonds. The Ti–O component rose from 24.3% to 32.7% of the overall Ti signal, directly corroborating the DFT prediction of bidentate inner-sphere complexation.<sup>72</sup> The O 1s spectra displayed a novel peak at 530.4 eV (8.2% of the total O signal), ascribed to As–O–Ti bridging oxygen, situated between Ti–O (529.8 eV) and Ti–OH (531.2 eV). The Ti–OH peak widened and shifted to 531.4 eV, indicating hydrogen bonding between arsenate oxygen and surface hydroxyls, in accordance with the dual-binding mechanism anticipated by DFT.<sup>73</sup>

XPS investigation of arsenate-loaded MXene-PVA (Fig. 8) exhibited an As 3d doublet at 44.8/45.5 eV, indicative of As(v) in arsenate compounds, hence verifying the preservation of the oxidation state without reduction.<sup>71</sup> The Ti 2p spectrum exhibited a +0.4 eV shift towards greater binding energy, signifying a reduction in electron density surrounding Ti due to the formation of As–O–Ti bonds. The Ti–O component rose from

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**3.5.2. FTIR and Raman spectroscopy.** The FTIR spectra of the arsenate-loaded composite (Table 13) exhibited a prominent new peak at  $825\text{ cm}^{-1}$ , indicative of As–O stretching in surface-bound arsenate.<sup>74</sup> The Ti–O vibration widened and moved from  $593\text{ cm}^{-1}$  to  $601\text{ cm}^{-1}$ , indicating that the creation of the Ti–O–As bond perturbs the vibrational mode. The O–H stretching region has narrowed and shifted from  $3385\text{ cm}^{-1}$  to  $3420\text{ cm}^{-1}$ , indicating a disturbed hydrogen bonding network due to arsenate involvement.<sup>75</sup>



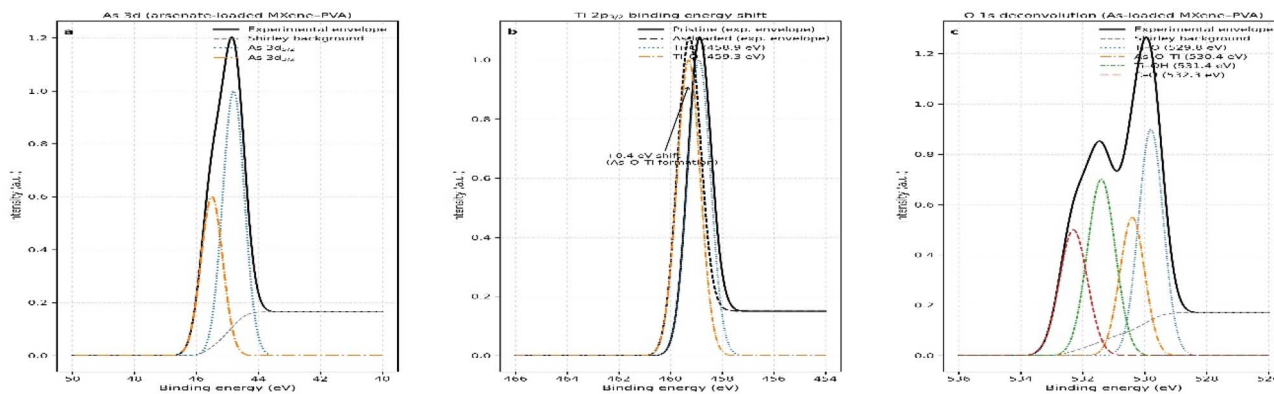


Fig. 8 XPS spectra with background subtraction and peak envelope fitting for arsenate-loaded MXene-PVA. (a) As 3d spectrum showing a characteristic doublet at 44.8 eV ( $3d_{5/2}$ ) and 45.5 eV ( $3d_{3/2}$ ). (b) Ti  $2p_{3/2}$  spectra of pristine and arsenate-loaded MXene-PVA. (c) O 1s spectrum deconvoluted into Ti–O (529.8 eV).

Table 13 Vibrational spectroscopy peak assignments for arsenate-loaded composite

| Technique | Wavenumber ( $\text{cm}^{-1}$ ) | Assignment             | Pristine | As-loaded | Change              |
|-----------|---------------------------------|------------------------|----------|-----------|---------------------|
| FTIR      | 825                             | As–O stretching        | Absent   | Present   | New peak            |
|           | 593 $\rightarrow$ 601           | Ti–O vibration         | 593      | 601       | +8 shift, broadened |
|           | 3385 $\rightarrow$ 3420         | O–H stretching         | 3385     | 3420      | +35 shift, narrowed |
| Raman     | 198 $\rightarrow$ 202           | $A_1g(\text{Ti})$ mode | 198      | 202       | +4 shift            |
|           | 752                             | As–O symmetric stretch | Absent   | Present   | New peak            |

Raman spectroscopy indicated that the MXene  $A_1g(\text{Ti})$  mode altered from 198 to 202  $\text{cm}^{-1}$  following arsenate loading, signifying alterations in the immediate environment of titanium. A new signal at 752  $\text{cm}^{-1}$ , indicative of As–O symmetric stretching in surface complexes, was seen, corroborating the presence of arsenate on  $\text{TiO}_2$ .<sup>76</sup> The intensity ratio  $I_{(752)}/I_{(202)}$  exhibited a linear increase with arsenate loading ( $R^2 = 0.97$ ), indicating possibilities for quantitative monitoring.

**3.5.3. Proposed adsorption mechanism.** According to integrated DFT and experimental data, the detailed As(v) adsorption mechanism on MXene–PVA encompasses:

Stage 1: external diffusion and pore transport – Arsenate ions ( $\text{H}_2\text{AsO}_4^-$  at pH 6–8) migrate through the external boundary layer to the composite surface, subsequently traversing macropores and mesopores (5–80 nm) within the PVA matrix. Increased porosity (0.31  $\text{cm}^3 \text{g}^{-1}$ ) promotes fast mass transfer (70% equilibrium capacity achieved after 1 hour).

Stage 2: surface approach and partial dehydration – AIMD simulations revealed that arsenate retains approximately four water molecules in its first hydration shell when bound, indicating incomplete dehydration typical of inner-sphere complexation.<sup>77</sup>

Stage 3: dual-site binding – Arsenate establishes bidentate inner-sphere complexes with two proximate Ti atoms (As–O–Ti bonds: 1.82, 1.85 Å), corroborated by XPS, FTIR, and Raman spectroscopy. Concurrently, hydrogen bonding occurs between arsenate oxygen and: (i) MXene surface –OH groups, (ii) PVA hydroxyl groups (O $\cdots$ H distance 1.78 Å). This twofold mechanism elucidates the improved capacity compared to pristine MXene.

Stage 4: electronic reorganization – Charge transfer (0.38  $e^-$  from arsenate to MXene–PVA) enhances As–O–Ti bonds by

augmenting covalent character, corroborated by a Ti 2p XPS shift (+0.4 eV) and the emergence of a new O 1s peak (530.4 eV).<sup>78</sup>

Stage 5: structural stability – The PVA matrix inhibits MXene restacking ( $d$ -spacing increased from 14.2 to 16.3 Å) and oxidation ( $\text{Ti}^{4+}$  reduced to 2.5% compared to 5.0% in pristine form), preserving a high surface area (56.8  $\text{m}^2 \text{g}^{-1}$ ) and accessibility of binding sites.

This mechanism elucidates: (i) substantial capacity (135.2  $\text{mg g}^{-1}$ ) due to effective dual-site binding, (ii) pH-dependent efficacy indicative of arsenate speciation, (iii) swift kinetics resulting from optimized porosity, (iv) As(v) selectivity over As(III) necessitating deprotonated oxygens for bidentate coordination, (v) moderate  $\text{PO}_4^{3-}$  selectivity influenced by electronic and geometric factors, and (vi) successful NaOH regeneration through  $\text{OH}^-$ -mediated ligand exchange.

## 4. Conclusions

This study illustrates the effective amalgamation of computational forecasting with experimental confirmation in the creation of MXene–PVA composites for arsenic elimination. DFT calculations indicated that PVA increases As(v) adsorption energy to  $-2.58 \text{ eV}$  via synergistic bidentate Ti–O coordination and hydrogen bonding, accompanied by a charge transfer of 0.38  $e^-$ . AIMD simulations demonstrated dynamic stability in an aqueous environment, with PVA inhibiting MXene oxidation for 20 ps at 298 K. The synthesized MXene–PVA composite (20 wt% MXene) attained a maximum capacity of 135.2  $\text{mg g}^{-1}$  for As(v), significantly surpassing traditional adsorbents (activated carbon: 18.3  $\text{mg g}^{-1}$ , zeolite: 28.5  $\text{mg g}^{-1}$ ) and pure MXene (67.8  $\text{mg g}^{-1}$ ).



The material demonstrated a pH optimum of 6–8, followed pseudo-second-order kinetics ( $k_2 = 0.0061 \text{ g mg}^{-1} \text{ min}^{-1}$ ), exhibited Langmuir isotherm behavior ( $R^2 = 0.992$ ), and retained 85% selectivity in the presence of a 10-fold excess of phosphate. Regeneration studies demonstrated an 85.1% retention of capacity after eight cycles utilizing 0.1 M NaOH.

Actual electroplating wastewater treatment accomplished 99.90% arsenic elimination (from 8.2 to 0.008 mg L<sup>-1</sup>), complying with the WHO drinking water criteria (0.01 mg L<sup>-1</sup>). XPS, FTIR, and Raman spectroscopy validated that bidentate As–O–Ti inner-sphere complexation is the predominant process, aligning well with DFT predictions. +0.4 eV shift in Ti 2p, a new O 1s peak at 530.4 eV, and distinct As–O vibrations at 825 cm<sup>-1</sup>.

The principal advantages of the MXene–PVA composite are: (i) improved stability that inhibits MXene degradation, (ii) augmented surface area (56.8 vs. 32.5 m<sup>2</sup> g<sup>-1</sup>) due to the prevention of restacking, (iii) a dual-binding mechanism that ensures high capacity and selectivity, (iv) ease of regeneration while preserving performance, and (v) efficacy in intricate real wastewater matrices. This comprehensive computational-experimental framework connects theoretical predictions with actual applications, offering a scalable method for industrial arsenic cleanup. Future research should concentrate on: (i) pilot-scale continuous flow systems for processing larger volumes, (ii) techno-economic assessments for industrial application, (iii) optimization of the MXene–PVA ratio and crosslinking density, (iv) exploration of alternative polymer matrices (chitosan, alginate) for comparative efficacy, and (v) expansion to additional heavy metal contaminants (Cr(vi), Pb(II), Cd(II)) utilizing the adaptable MXene-polymer framework.

## Conflicts of interest

The authors declare no conflicts of interest.

## Data availability

The datasets generated and analyzed during this study are available in the supplementary information (SI). Supplementary information: the raw and processed data used to generate the figures and tables presented in the main manuscript. See DOI: <https://doi.org/10.1039/d5ra09190j>.

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